8
Linguistic Scaffolding and Scientific Realism

> A deformable body, as distinguished from a rigid body, is one which is susceptible of a change of form either as a whole or in any of its parts. Strictly speaking, all bodies in Nature are deformable, for there are no motions which are not accompanied by greater or smaller changes of form in the bodies participating in them. But in many cases—for example, in that of the pendulum, the lever, the top—it is sufficient, for a first approximation to reality, to assume the bodies in question as rigid. . . . Nature does not allow herself to be exhaustively expressed in human thought.
>
> Max Planck (Planck 1932, 1–2)

(i)

At a certain point in his discussion, Heinrich Hertz explicitly disassociates his own philosophical opinions from the more militant forms of anti-metaphysical opposition that he saw in fellow philosopher/physicists such as Ernst Mach:

> The usual answer, which physics nowadays keeps ready for such attacks, is that these considerations are based upon metaphysical assumptions; that physics has renounced these, and no longer recognizes it as its duty to meet the demands of metaphysics. . . . If we had to decide upon such a matter, we should not think it unfair to place ourselves rather on the side of the attack than of the defense. A doubt which makes an impression on our mind cannot be removed by calling it metaphysical; every thoughtful mind as such has needs which scientific men are accustomed to denote as metaphysical. Moreover, in the case in question, as indeed in all others, it is possible to show what are the sound and just sources of our needs. (Hertz 1894, 23)

Hertz does not amplify his concerns fully, but he appears to have in mind some vital "small metaphysics" considerations that have become almost completely obliterated within the cruder, post-Carnapian conception of how an opposition

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to "metaphysics" should be framed.[1] My central purpose in this penultimate chapter is to elevate these suppressed "small" concerns to philosophical attention once again. Perhaps surprisingly, Hertz begins his discussion by declaring that the first part of *Principles* should be regarded as synthetic a priori in a Kantian vein throughout, with no direct reliance upon empirical behavior at all:

> The subject-matter of [our] first book is completely independent of experience. All the assertions made are a priori judgments in Kant's sense. They are based upon the laws of the internal intuition of, and upon the logical forms followed by, the person who makes the assertions; with his external experience they have no other connection than these intuitions and forms may have. (Hertz 1894, 45)

These remarks are notoriously cryptic (Lützen 2005, 121–3), but Hertz clearly hopes to assemble a specific realm of firm objective data to which physics should be held responsible. In doing so, we should follow the reasoning procedures which strike us as proving the best guarantors of deductive soundness, which Hertz identifies with guiding principles of traditional Euclidean geometry (in company with Kant). But this merely represents a jumping off point, for Hertz also embraces a two-tiered methodological approach in which enterprising physicists remain free to follow their creative hunches into significantly distinct "extension realms," like the projective realm that had been devised by projective geometers earlier in the century (in which Euclid became supplemented with large quantities of totally unintuitive "extension points") (Wilson 2020). Hertz himself engages in a large amount of rather outrageous supplementation of this general character, as outlined in the appendix to Chapter 3.[2] But it is only within this enhanced realm that Hertz finds the platform upon which his hedged form of scientific realism ultimately rests, for "the totality of things visible and tangible do not form a

[1] The specific "metaphysical" objection he raises applies to employing Hamilton's principle as a "second image" foundational doctrine on the grounds that it "attributes intentions to inanimate nature" (Hertz 1894, 23). As indicated previously, he expands this criticism by pointing out that such tactics pick out the wrong behaviors in the case of non-holonomic conditions (choosing paths of optimal control rather than the proper frictionless response—see Bloch (2003)). Hertz's discussion here is carefully and admirably observed, and I have been surprised to find later commentators loosely presuming that Hertz regards his three "images" as descriptively equivalent. But it is also common for contemporary philosophers to loosely conclude that various physical doctrines are "fully equivalent" to one another on the grounds that they coincide within a purist point mass setting of the "Newtonian" sort discussed in Chapter 3.

[2] He appears to assume that we should follow Kant's Euclidean geometrical percepts in constructing the base-level properties of the interlocked mechanisms we find in nature, viz., the ways in which their parts connect with one another and the freedoms of movement they display as a connected ensemble. These geometrical constructions then characterize the "mobility space" facts that comprise the prime data to which Hertz's system of mechanics is ultimately responsible. The reliability of these constructions is fully guaranteed by the a priori principles we employ in ascertaining these basic mobilities. However, to frame a universe governed by simple and comprehensive laws. we must enrich these base-level descriptions into becoming the extended universes described in the appendix to Chapter 3 (i.e., as the manifolds $\mathcal{M}$ that cover the mobility spaces S).

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universe conformable to law" (Hertz 1894, 25–6). Indeed, the empirical "law" that ultimately governs dynamic change within Hertz's hypothetically enlarged realm appeals to movements within a high dimensional manifold bearing a metric of a decidedly non-Euclidean character. Hertz clearly believes that a properly conceived "metaphysics" should not prohibit the ingenious scientist from proposing radical reappraisals of this type.[3]

This is where Hertz's appeal to a parallelism between "images in thought" and "consequences in nature" enters the scene; it provides the bridge whereby he reaches out to the external world from the framework assembled in the *Principles*. He acquired this minimalist recipe for learning about "things in themselves" from his teacher Helmholtz, who wrote in a celebrated passage:

> Natural science . . . seeks to separate off that which is definition, symbolism, representational form or hypothesis, in order to have left over unalloyed what belongs to the world of actuality whose laws it seeks. . . . The relation between the two of them is restricted to the fact that like objects exerting an influence under like circumstances evoke like signs, and that therefore unlike signs always correspond to unlike influences. To popular opinion, which accepts in good faith that the images which our senses give us of things are wholly true, this residue of similarity acknowledged by us may seem very trivial. In fact, it is not trivial. For with it one can still achieve something of the greatest importance, namely forming an image of lawfulness in the processes of the actual world. Every law of nature asserts that upon preconditions alike in a certain respect, there always follow consequences that are alike in a certain other respect. Since like things are indicated in our world of sensation by like signs, an equally regular sequence will also correspond in the domain of our sensations to the sequence of like effects by law of nature upon like causes. . . . Thus although our sensations, as regards their quality, are only signs whose particular character depends wholly upon our own makeup, they are still not to be dismissed as a mere semblance, but they are precisely signs of something, be it something existing or happening, and—what is most important—they can form for us an image of the law of this thing which is happening. (Helmholtz 1878, 118)

As we noted in our Chapter 5 criticisms of Hertz's "images in thought/ consequences in nature" passage, these simple parallelisms are rarely valid, and the facts of objective nature must be determined from our pragmatically rewarded procedures in other ways, in a manner to which I will later return.

[3] Scientific realists are generally inclined to reject the "world of familiar acquaintance" themes of Stebbing (1937) (see also Wilson (2017, 40–6)). It was common in Hertz's era to presume that free-spirited human creativity represents a mental capacity superior to the allegedly "constant conjunction" character of animal thinking (Wilson 2020, 66–7). Whether Hertz subscribes to such themes or not, I cannot certify.

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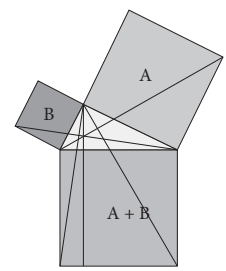
We should observe that such doctrines rely heavily upon the traditional presumption that we are intimately and fully acquainted with the evanescent qualities of our mental experiences (Hertz's "images"), but the internal feature of the objective properties of the external world remain permanently beyond our intellectual grasp. Within this wintery assessment, all we can ever learn about the external world is supplied within the behavioral husks that we can glean from the Harpo-imitates-Groucho parallelisms upon which science depends. This dichotomy between our allegedly robust "grasp" on the "mental" and our feeble acquaintance with the "physical" remains with us to this day, although the better angels of twentieth-century philosophy vigorously attempted to reject these unfortunate motifs of "private language" thinking. I will merely comment that I believe that these critics are right.

(ii)

However we correct these misguided doctrines with respect to conceptual "grasp," a fundamental quandary persists. To frame a cogent portrait of external circumstance, we must stoutly rely upon the principles of mathematical reasoning that we regard as inherently the most sound and reliable. But these same reasoning tools may incorporate interior ingredients that we will not wish to foist upon the external world. Hertz's appeals to the securities of Euclidean construction apparently trace to the fact that these tactics undergird our firmest advances into the inferential unknown. But they frequently do so in a manner that may strike us as rather mysterious from a psychological point of view. Consider the Pythagorean theorem. Staring blankly at a right triangle, we may idly frame the hunch that the squares on its sides add up in a Pythagorean manner, but this is not an ironclad proof. But imbed this figure within an appropriate nest of supplementary lines, and our triangle becomes entrapped within a mathematical box that leaves the figure no option except to prove Pythagorean. Clearly deep mathematics gains many of its ampliative powers through more radical forms of supplementary embedding. Within an unanticipated richer setting, internal relationships that had heretofore escaped our notice may become vibrantly salient (Kant characterizes this surprising source of supplementary certainty as the "synthetic a priori"). Unlike Mach, Hertz does not want to adopt raw sense data as the base domain to which science is ultimately responsible, but to instead utilize the geometricized constructions that Kant tells us how to construct via Euclidean principles (it is here that we obtain the machine-like "mobilities" discussed in Chapter 4).



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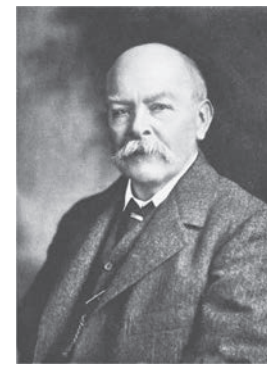
But this is where a new dilemma intrudes. Given Hertz's strong reliance upon intermediate ampliative reasoning of this Euclidean character, how can we coherently understand his contention that, for all that, the external world itself might be credited with a geometry of a markedly different character (or, possibly, no certifiable geometry at all)? A few years after Hertz's *Principles*, the physicist J. H. Poynting evocatively captures these methodological concerns as follows:

> While the building of nature is growing spontaneously from within, the model of it we seek to construct in our descriptive science, can only be constructed by means of scaffolding from without, a scaffolding of hypotheses. While in the real building all is continuous, in our model there are detached parts, which must be connected with the rest by temporary ladders and passages, or which must be supported till we can see how to fill in the understructure. To give the hypotheses equal validity with the facts is to confuse the temporary scaffolding with the building itself.
>
> (Poynting 1899, 385)



Poynting

Some of the "temporary scaffoldings" that Poynting appears to have in mind are the roundabout but seemingly necessary "foundational" appeals to point masses or rigid bodies to which Victorian physicists first appealed in their attempts to reach an exterior universe comprised of fields and the other forms of continuous media capable of supporting wavelike processes. In the appendix to Chapter 3, I outline the manner in which these nineteenth century appeals to "essential idealization" were later addressed with the direct mathematics of limits and integral equations (these replacement pathways were suggested by Hilbert in his 6th problem on mechanics). Despite these improvements, the residual worries of a Poynting remain: in framing our portraits of the external world in mathematical terms, at which point do we become free to throw away the unwanted features that may accompany these constructions? Joseph Boussinesq writes, in advancing a standard Kantian theme:

> The human mind while observing natural phenomenon recognizes in them, besides many clear elements which it does not get to unravel, one clear element, which by virtue of its precision is liable to be the object of truly scientific knowledge. That element is the geometric one, pertaining to the localization of objects in space which permits one to represent them, to draw them, or to construct them in a more or less ideal manner. (Boussinesq 1880, 88)

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But if we don't regard these human "precisions" as fully representative of external reality, how should we proceed? This is exactly the dilemma that Hertz confronts in his two-tiered approach to satisfactory modeling (viz., he claims that all of the conclusions reached in the first half of *Principles* are thoroughly Euclidean in their mathematical reliability, yet in his "empirical" follow up he abandons that preliminary scaffolding in favor of higher-dimensional manifolds possessing a non-Euclidean metric). But it is only at this second stage of construction that Hertz finds the altered stream of mathematical "images" that he hopes can successfully parallel the objective structures present in external reality. But if we must first climb up a Euclidian ladder to reach a decidedly non-Euclidean perch, upon what basis are we allowed to throw our initial scaffolding away in reaching a truer estimation of the strange new world that lies beyond?

> Scientific accuracy requires of us that we should in no wise confuse the simple and homely figure, as it is presented to us by nature, with the gay garment which we use to clothe it. Of our own free will, we can make no change whatever in the form of the one, but the cut and color of the other we can choose as we please.
> (Hertz 1893, 28)

At this point a number of "small metaphysics" questions pertaining to the reliability of our mathematical capacities make their entrance. What degree of similitude can we expect will hold between the streams of calculation that we are able to certify as mathematically rigorous and the behavioral expectations that we hope to attribute on this basis to the world beyond?

Mach's brutal response to Poynting's worries with respect to excessive scaffolding maintains that we needn't worry about any hypothesized external world at all, beyond a pure phenomenalism of a neutral monist complexion. Like most of us, Hertz would prefer avoiding these anti-realist extremes without embracing any of the otherwise intrusive metaphysical demands that Mach rightly rejects. Hertz accordingly decides that some residual dose of smaller-bore "metaphysics" is needed to address Poynting's concerns with respect to "scaffolding." And my own assessment of our intellectual circumstances remains roughly the same.

What is the striking fact about Kantian thought that appears to have particularly impressed Hertz? Most likely it resides in the ampliative certitude that we mysteriously gain from following the strict constructive pathways that Euclid outlines. Even if we eventually decide to extend our constructions beyond this point in Hertz's "second part" manner, we recognize that their descriptive merits derive entirely from the ways in which they simplify the Euclidean basis upon which they are built. But similar crablike advancements in inferential trustworthiness affect every portion of effective applied mathematics. Euler mistrusted Daniel Bernoulli's hunch that arbitrary positions of a violin string could be spectrally decomposed into sine wave factors, but this factorization capacity was

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mathematically ratified later by Fourier, Dirichlet, and Sturm and Liouville (thereby supplying nineteenth-century physics with the vital "poem" (Thomson and Tait 1912, 470)) that underwrote many of the subsequent advancements achieved in the over half century that follows. But the security of Sturm and Liouville's advancements is ratified only by embedding the target problems within a set theoretical netting of a decidedly modernist cast (Wilson 2017, 399–402).

The stepwise character of intellectual advance stems directly from our inherently limited reasoning skills that we must somehow transcend through clever stratagems and wiles. We merely need to remember, as Mach constantly stresses, the fact that evolutionary process has fashioned us with computational capacities better suited to the homey tasks of securing food and shelter than unraveling the exterior mysteries of light and matter. We collectively face a methodological dilemma similar to the one that Descartes confronts in his *Sixth Meditation* (Descartes 1641): how can we ascertain the true features of the exterior world, given that we must largely depend upon sensory signals that have been overtly fashioned to keep our bodies safe and healthy rather than becoming objectively knowledgeable? Descartes's striking answer appeals to reverse-engineering the reasons why we possess the sensory systems we do, by considering how limited creatures such as ourselves might plausibly deal with the complicated environments in which we find ourselves. From this perspective we find that the sensory signals we intuitively regard as "simple" actually represent highly processed responses[4] to external promptings that allow human beings to address their needs in swift and generally effective ways. The better we can appreciate the reasons why our sensory skills are constructed in this biased-in-favor-of-human-capacity manner, the more clearly we will recognize what the objective world is like in its own right. These Cartesian doctrines strike me as quintessentially "scientific realist" in character, and our eventual response to Poynting's challenge will follow a similar pattern.[5]

Quite generally, nineteenth-century scientific thought reflects the methodological realization that great strides forward in effectively reasoning can be achieved by embedding initial doctrines within richer extended settings that allow (1) for more effective reasoning patterns and (2) perhaps capture external reality better. The unexpected recasting of traditional Euclid ean doctrine at the hands of the projective geometers possibly provides the central paradigm of these activities, but enlarged algebraic conceptions of number-like objects were prompted by analogous concerns (Wilson 2020). We have already seen that

[4] Descartes labels these notions as "confused ideas," whereas Chapter 5 would consider them as "suitably homogenized."

[5] The central differences lie in the fact that a benevolent deity supplies Descartes with the diagnostic tools required to unravel this engineering upon an a priori basis, whereas we are obliged to work with a ragtag toolkit that we have gradually fashioned for ourselves based upon ancestral inheritance and a large degree of trial-and-error experimentation.

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Hertz's own proclivities reflect both of these methodological tenets, there explored, perhaps, to an excessively speculative degree.

Of these two objectives, Richard Dedekind maintained that mathematicians' central motivational duty is to task (1), not (2). In studying a given stretch of reasoning, they should explore its wider neighborhood inside mathematics fully, looking for nearby but improved reasoning opportunities. Doing so often requires a wider overview of a very large catalog of potential reasoning domains. It is in this spirit that Dedekind himself explored the circumstances in which a starting algebra could be extended to a wider frame in which unique factorization is restored. Likewise, mathematicians should investigate what happens when we apply differential equation-like reasoning to complex-valued functions, to stochastic processes, to fractional derivatives, to fractals, etc., even if their motivating interests seem to apply to real-valued domains only (in physics, say). Only within those enlarged settings will we sometimes find the *guideposts* we require to plot our reasoning strategies successfully.

Why is this so? These bootstrapping capacities for exploring wider applicational realms and cobbling together corrected forms of older reasoning patterns represent some of the chief tactics we employ for transcending the parochial limitations on reasoning that we have inherited from our hunter-gatherer ancestors. Even today, we adjudicate the prima facie trustworthiness of a computer program by determining how it performs over a familiar applicational domain, before it is applied to the topics for which it was hopefully designed. Such a check on pragmatic "reliability" is scarcely foolproof, but it is clearly helpful. To be sure, more sophisticated examinations of soundness can supply tighter assurances, but they are frequently hard to construct.

(iii)

Such considerations bring us face to face with another downstream consequence of Carnapian thinking that has seriously confused philosophical thinking on these issues ever since. I have in mind Quine's celebrated criterion of ontological commitment (Quine 1953, 1–19): to determine whether an agent is committed to a range of objects O, force her to articulate her beliefs, as best she can, into the uniform format of a first-order theory T and scrutinize the existential claims she is logically forced to accept, looking for "there is an x such that x is an O." Quine calls this hypothetical process of adopting these theory T proprieties "regimentation" and regards it as a matter of simple intellectual honesty (the following is a paraphrase, not an actual Quine quote):

> If you don't believe in the reality of an ether, witches, or numbers, stop saying things like "there is an ether through which transverse waves travel" and "there is

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> a witch x that caused a lot of trouble in old Salem." Find some alternative means of expressing the salvable core of what you wish to express within a properly uniformized format. Perhaps these regimented recastings might become "there was a person who was called a 'witch' that occasioned a lot of trouble back in old Salem" and "there are electromagnetic waves which are purely transversal, but there is no ether through which they travel." But the folks who cavalierly claim that "there is an ω-sequence of sets that can serve as a surrogate for the natural numbers" will find it harder to paraphrase away their apparent commitments to a range of Platonic entities. Carnap's contention that such assertions merely "represent 'framework principles'" does not represent an adequately honest answer.

Quine's arguments on this last score appeal to the same gradualist presumptions with respect to linguistic development that we have highlighted. A society may originally presume that "F = ma" represents a framework cornerstone of its physical thinking, but subtle adjustments in profitable modeling technique can dislodge it from that former centrality.

What I find odd about Quine's doctrines lies in their off-handed presumptions that "regimenting" the things we claim in everyday practice represents a comparatively easy task and we can safely assume (as a "convenient philosophical fiction") that it has been successfully accomplished.[6] But this is not so. Our improving standards of practical achievement reflect the inconvenient fact that, in coming to scientific grips with a recalcitrant physical world, we must cobble together the limited collection of sensory and reasoning abilities that we have inherited from our ancestors into novel combinations that allow us to escape from the narrow confines of practicality for which these skills had originally evolved. We have further learned that through exploiting mathematical thinking of a rather purist cast, we can extend and patch together these originally limited resources to more widely varied and reliable purposes. And fresh warnings and refinements with respect to these beacons of reliable reasoning continue to emerge. The early geometers learned that they could strongly certify the correctness of relationships such as the Pythagorean theorem by adding supplementary lines to the figure and by subdividing and moving around its component areas. But the mathematicians of antiquity eventually realized (presumably on the basis of harsh negative experience) that the latter techniques sometimes led to erroneous conclusions (such as all triangles are equilateral). In this fashion were Euclid's tight strictures on constructive proof discovered. But why, exactly, do these two policies of proof so differ in their reliability conferring powers? Kant believed that he had found a philosophical answer, but recent

[6] Better, I think, to recall the old folk song and regard "theories" as "young, but daily growing."

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studies in the representational capacities of geometric diagrams have helped address these questions in a more satisfactory manner, in conjunction with a better delineation of our native cognitive resources.

In the same way, we understand (because he wrote about them extensively) the policies of quasi-experimentation that led Oliver Heaviside to his remarkable techniques for extracting extremely valuable and reliable data from linear differential equations. But the underling sources of their deductive surety remained mysterious for many years (in his own assessment, "in reality the logic (of it) is the very last thing" (Heaviside 1912, III 370). In developmental contexts such as these, we must frequently accept as a brute empirical fact that certain veins of mathematicized enlargement and contrast often provide better pathways to reliable conclusion than others, but we may remain quite unsure of what we find ourselves "committed to" in a Quinean vein.

As the central core of this book has argued, our successful multiscalar practices within classical physics likewise resist ready "regimentation" into a single-leveled theory T of the sort that Quine demands. Why these patched together tactics work as well as they do frequently require strategic assessments of a very sophisticated order. There needs to be a "logic" in Heaviside's sense as to why this happens, but teasing it out can prove immensely difficult. But this shouldn't seem surprising. As we previously observed, familiar feats in "mental telepathy" trade upon the art of extracting enough information from a selected mark X to deductively establish that A holds of X without the innocent victim realizing that she has supplied as much. But the magician who executes the subterfuge can easily learn how to apply the "information sieve" that extracts A from the data provided without himself possessing any strategic understanding of how the "sieve" manages to process the information received (Wilson 2017, 389–91). In such circumstances, a sensible response to Poynting's concerns with respect to "scaffolding removal" demands answers of a considerably different character than Quine encourages.

In the previous chapter, we observed that Theodore Sider regards his prescientific approach to metaphysics as a direct consequence of his self-proclaimed "scientific realism." But this conception of "realism" relies upon the same implausible assumptions of theory T reformulation as Quine does. We'll return to Sider's Panglossian suppositions later in the chapter, after we first review the underlying conception of "scientific realism."

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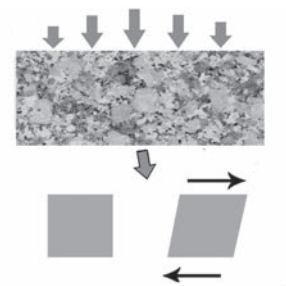
(iv)

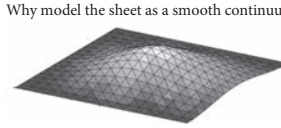
Let us first observe that no reasonable conception of "scientific realism" should entangle itself in syntactic requirements of a Quinean character. This chapter begins with a quotation from the physicist Max Planck who is usually regarded as one of the great defenders of scientific realism (he engaged in a famous debate with Ernst Mach on the topic in 1910–11). But he freely acknowledges that "nature does not allow herself to be exhaustively expressed in human thought" (Planck 1932, 1–2), where by "human thought," he means "in mathematized terms." On this understanding, Planck is merely noting the undeniable fact that our successful policies of mathematical registration generally ignore potentially disruptive lower degrees of freedom until such time as explicit attention to their fine-grained complexities is required. As these subdominant disturbances become salient, scientists shift their attention to altered models better suited to these suppressed factors. The only demand that Planck's "realism" places upon these later adjustments appears in the form of the "important and inevitable postulate" that:

> the different hypotheses introduced for dealing with different problems should be compatible with each other. For otherwise the physical picture of the world would lose its uniformity, and in some circumstances we should get two mutually contradictory answers to a definite question. (Planck 1932, 1–2)

The homogenization barriers characteristic of a standard multiscalar architecture provide an unanticipated resolution to Planck's concerns, for they guard against the emergence of "mutually contradictory answers" without mandating descriptive complexity of an unmanageable character. By working together in cooperative harmony, a suitably monitored patchwork of submodels can capture a wide range of physical behaviors quite admirably without demanding that any single modeling codify all of the target system's varied behaviors within a united and amalgamated narrative. By keeping its "dominant behavior" conclusions suitably stratified, the explosions within descriptive variables characteristic of Tao's "curse of dimensionality" can be avoided.



Why model the sheet as a smooth continuum?



Isn't a finite element mesh more realistic?

Pursuing these considerations a bit further, we can provide a helpful response to Poynting's challenge. As noted in the previous chapter, multiscalar architectures suitable for complex materials often utilize smooth continuum mechanics as the formalism of choice for capturing behaviors upon

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the higher lengths ΔL. The architecture shifts to a subdivided laminate submodeling only as exceptional events involving lower-scale grain become pertinent to the ΔL-level behaviors. But why should we engage in this continuum physics pretense, given that we are fully aware of the fact that our material is not truly smooth in the manner indicated? Ludwig Boltzmann complains about this artificial descriptive smoothing in his celebrated defense of "atomism":

> The differential equations of mathematico-physical phenomenology are evidently nothing but rules for forming and combining numbers and geometrical concepts, and these in turn are nothing but mental pictures from which appearances can be predicted. . . . Mathematico-physical phenomenology sometimes combines giving preference to the equations with a certain disdain for atomism. . . . [D]efinite integrals that represent the solution of the differential equation can in general be calculated only by mechanical quadratures and thus again demand division into a finite number of parts. Do not imagine that by means of the word continuum or the writing down of a differential equation, you have acquired a clear concept of the continuum. On closer scrutiny the differential equation is merely the expression for the fact that one must first imagine a finite number; this is the first prerequisite, only then is the number to grow until its further growth has no further influence. What is the use of concealing the requirement of imagining a large number of individuals now, when at the stage of explaining the differential equation one has used that requirement to define the value expressed by that number?
>
> If I then declare differential equations, or a formula containing definite integrals, to be the most appropriate picture, I surrender to an illusion if I imagine that I have thus banished atomistic conceptions from my mental pictures.
>
> (Boltzmann 1897, 42–3)

What Boltzmann has in mind is this. Consider a sheet of steel, which under a microscope appears as an elaborate warren of crystalline grains. Won't modeling the metal as a numerical array of descriptive units of finite length ΔL remain truer to reality than pretending the sheet is smoothly continuous all of the way down to the infinitesimal-size scale? Shouldn't we accordingly view the numerical "finite element" grid at the bottom of the illustration as more "realistic" than the smoothed-over continuum described in the differential equations that we employ at this stage of our multiscalar modeling?

The methodological naïveté inherent in this passage is astonishing and plainly offers no real defense of atomism at all. But the example returns us to the importance of the *guideposts to modeling reliability* that we discussed in section (iii). Whenever we reasonably can, we should strive to capture a physical behavior within the net of smooth differential equations because the latter represents the mathematical landscape in which we most readily find the clues that warn us that

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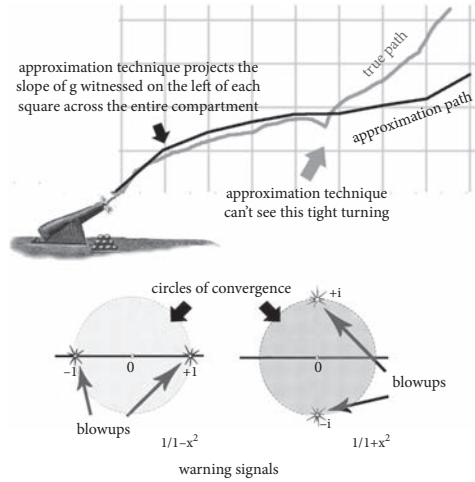

a numerical computation upon a computer may stray from a salutatory path. As we noted in Chapter 5 (iii), it is only at the continuous function level that we can detect the tiny blip that can push a computer simulation into serious inaccuracy. If we remain steadfastly at the level of Boltzmann's "finite division of parts" (i.e., a finite difference scheme of resolution length Δx), we will overlook this warning indicator and deprive ourselves of the helpful signposts of inferential reward and hazard found only at the infinitesimal level of the differential equations. Similar warnings apply to series summations. Computers don't mind adding up whatever factors we give them, no matter whether the outcomes are descriptively nonsensical or not. But the standard indicators of summation failure often lie in the singularities located upon the complex plane outside of the real line (Wilson 2017, 393–4). Indeed, a computer can't distinguish a true infinitary blowup from a really big number unless we somehow supply it with guiding criteria. But the "stop" and "go" signals required to monitor the trustworthiness of a simple series summation commonly require expeditions into further realms of faraway structure, sometimes of a quite exotic stripe.[7]

A subtler version of this same misunderstanding is found in Poston and Stewart (1978, 300), with respect to the subtle functional analysis considerations one now finds in a sophisticated engineering primer:

> We have gone beyond the casual assumption of "configurations pretty much like smooth functions if we don't look too closely"....We are making very delicate hypotheses (like square integrable second distributional derivatives) about exactly what we would see if we could look arbitrarily close: hypotheses we know to be false. In what sense these "infinitesimal" hypotheses can be said to model the local average nature of elastic solids is a dark and gleaming mystery. The results of the model, in macroscopic predictions, are very successful indeed; but it is hard to say in what sense the theory is more exact than a computer

[7] From this point of view, blow-up singularities on the complex plane frequently supply us with commonly represent our most effective instruments of physical data registration. See Wilson (2006) for more on this theme.

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> treatment of a finite element model like that of section 4, but with many more rods and strings. . . . Long calculations that you do not understand, whether by number crunching or Banach space methods, will land you with bridges that fall down. A theory is only as good as your understanding of it.

I'll supply a fuller rejoinder below, but the crucial guidance required to construct a reliable computer program for estimating the internal stress within a notched metal plate directly depends upon the delicate features of the "square integrable second distributional derivatives" of which Poston and Stewart complain. Once again we must peek into a rather daunting corner of Greater Mathematicsland to find the practical guidance we require.

In a similar vein, I have sometimes heard critics wonder, "Why should we fuss so much about whether the Navier-Stokes equations (which represent a severely smoothed over continuum account of fluid behavior) genuinely support a transition to turbulence? Water, after all, is decidedly molecular in composition." A rigorous positive answer (which is generally regarded as one of the greatest outstanding problems within applied mathematics) will provide us with an abundance of useful information with respect to questions as to whether turbulence can be usefully captured, say, in terms of the speculative energy cascades posited in Chapter 6. Our abilities to successfully capture higher-scale dominant behaviors in manageable mathematical terms strongly dictates how we might cobble together a descriptive architecture responsive to its richer complexities.

Attention to these naturalistic considerations strikes me as essential to a just appraisal of what I like to call "our computational place in nature," viz., in view of the physical environments in which we live and the basic computational skills with which we find ourselves supplied, which descriptive tasks will we find ourselves able to execute and which not? And which foreseeable avenues of improvement still remain open to us? Quasi-biological considerations of this type comprise the primary topics of Wilson (2017, 30–2).

Pace Kantian contentions otherwise, these observations with respect to the palpable imbalances that characterize human inferential capacity may simply reflect inherited limitations upon our biological capacities for ampliative reasoning, similar to the factors that inhibit a lungfish's navigational capacities when it first clambers upon the land (I believe that Mach would have largely concurred in this opinion). But, like that intrepid ichthyological adventurer, we can gradually improve our practical capacities. By employing clever mix-or-match reasoning tactics, we can patiently work our way past many of our initial limitations in

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reasoning, leading to an improved and nuanced appraisal of our computational circumstances, along both favorable and unfavorable directions (we now recognize that we can successfully bisect an angle with compass and ruler, but not obtain a trisection, but our hunter-gatherer ancestors could not have grasped this fact so firmly). Perhaps the angels needn't reason in this circumscribed manner, but we must. Those empyrean entities haven't arisen as the raw products of natural evolution, but we have, and we must live with the limited abilities we've received.

None of this indicates that the availability of reliable numerical methods (such as a finite element computation) isn't important to multiscalar modeling as well. Indeed, it is only through such means that we will normally be able to estimate localized stresses within a loaded metal sheet of a minimally complicated geometry. It is merely that the analytic clues found within their motivating set of differential equations are needed to adjudicate the soundness of the numerical scheme itself (which is notoriously capable of cheerily computing completely spurious results if left unmonitored).

(v)

These last considerations point to an important issue that becomes utterly muddled through Quine's insistence upon amalgamated regimentation. These confusions arise from the fact that advancing large swatches of mathematics effectively, our perceived interests in their "references" typically shift to a significantly different basin of concern. Richard Dedekind explicates this divergence as follows:

> If one finds the chief goal of each science to be the endeavor to fathom the *truth*—that is, the truth which is either wholly external to us, or which, if it is related to us, is not our arbitrary creation, but a necessity independent of our activity—then one declares the final results, the final goal (to which one can in any case usually only approach) to be invariable, to be unchangeable. In contrast, science itself, which represents the course of human knowledge up to these results, is infinitely manifold, is capable of infinitely different representations. This is because, as the work of man, science is subject to his arbitrariness and to all the imperfections of his mental powers. . . . The introduction of such a [computational] concept as a *motif* for the arrangement of the system is, as it were, an hypothesis which one puts to the inner nature of the science; only in further development does the science answer; the greater or lesser *effectiveness* of such a concept determines its worth or worthlessness. (Dedekind 1854, 755–6)

As we've just seen, the most effective signposts to reasoning improvement and correction frequently lie outside the mathematical arena in which one originally begins (within physics, for example). Dedekind emphasizes the importance of the

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wide-ranging examination of reasoning "motifs" across a vast array of coherent settings, whether or not they directly bear upon any physical application or not. Quite frequently, these novel domains are first reached through some form of appeal to what Dedekind calls "persistence of theorems," the recognition that superior reasoning policies can be obtained if the irregularities that block deduction within previously conceived arenas can be removed through supplementary "ideal" ingredients.[8] Nevertheless, the terminologies that we retain in pursuing these exploratory inquiries can sometimes prove problematic, for subtle difficulties sometimes emerge when linguistic expressions that superficially appeared trustworthy fail to find adequate supports within the newly extended setting (the simplest example of such a referential failure is "6/0," but Riemann's unchecked appeals to "minimal surfaces" singlehandedly generated the greatest single crisis in nineteenth-century mathematics: the failure of the Dirichlet Principle) (Monna 1975). As a result, a mathematician will understandably appropriate the managerial utilities of "refers" to her own investigative purposes, despite the fact that any significant concern with their direct real-world support will have been left far behind. It is exactly in this context that Richard Courant highlights the parochial importance of "mathematical existence":

> But empirical evidence can never establish mathematical existence—nor can the mathematician's demand for existence be dismissed by the physicist as useless rigor. (Only a mathematical existence proof can ensure that the mathematical description of a physical phenomenon is meaningful.) (Courant 1950, 3)

However, as our attempts to adjudicate the merits of our deductive practices within the "mixed modalities" characteristic of applied mathematics research, both claimants on the notion of "reference" can naturally come into play (e.g., "Jan Mikusiński ably established mathematical referents for the inverse operations within the Heaviside calculus, but it's dubious that he thereby captured their physical referents in an entirely happy way. Schwartz's distributions are closer to the mark in those regards").

Quine's insistence upon unified regimentation immediately blurs these distinct purposes with respect to "existence" into an unhelpful muddle. Penelope Maddy has criticized these significant shifts in focal concern with respect to "reference" in a detailed manner, although sometimes in terminologies that I regard as suboptimal (Maddy 2007; Maddy 2020), (I prefer Dedekind's emphasis upon mathematics' interest in exploring various "motifs for the arrangement of a system")

[8] Two of Dedekind's motivational paradigms are the projective revolution within Euclidean geometry and Kummer's introduction of "ideal factors" into algebraic number theory. He then attempts to build up the familiar number systems utilizing analogous tactics (Wilson 2020).

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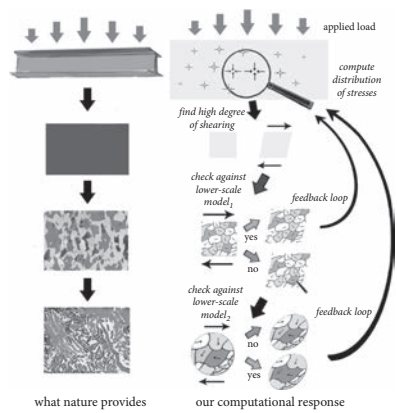
(Dedekind 1854, 756).[9] From this point of view, set theory gains its importance not by providing an all-embracing "ontology" for mathematics but as a set of extremely valuable tools for investigating the internal consistency of various mathematical "motifs" and their potential embeddings within one another (Wilson 2020).

These observations deepen our appreciation of why brute "constant conjunction" approaches to causation fail to capture the superior appeal of the smoother differential equation treatments that du Châtelet favors, albeit in a less metaphysically laden manner. We can still applaud her perceptive emphasis upon the importance of smooth differential equation behavior, even when some of this "importance" stems from the surprising manner in which the infinitesimal realm in which these smooth curves mathematically dwell provides the guideposts that humans require to reason reliably about the dominant behaviors of higher scale ensembles.

(vi)

These developmental considerations suggest that a suitable "realist" reply to Poynting's concerns with respect to "scaffolding" can be provided in a "reverse engineering" mode much like Descartes's. If we consider a multiscalar architecture suitable to a piece of steel, we can fairly easily distinguish the "guideposts to effective reasoning" aspects of the scheme from the natural behavioral opportunities to which they respond. Such divisions are common within any form of biomechanical explanation: we can distinguish the behavioral capacities of a colony of honeybees from the opportunities inherent in the pollen distributions that they seek to exploit. The fact that the shearing behavior of a moderately sized steel specimen is governed on a large scale by two bulk parameters and is able to largely regain its original shape when the applied force is removed represents a descriptive opportunity that resides within the metal itself. In contrast, the fact that we reason most effectively about these behaviors using smooth



[9] In particular, the terminologies that Maddy employs to this end ("arealism" and "thin realism") strike me as unhappily suggestive of a deflationist literature with which I have little sympathy. She wishes to distance her views from these contentions as well (Maddy 2022), but I worry that unwanted connotations linger on in her formulations. Possibly I'm overly sensitive on these scores, for standard deflationists trivialize the very investigations into word-to-world relationships that I regard as central to a "scientific realism" properly conceived. Semantic minimization of this character is utter anathema to me.

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continuum physics models significantly reflects the internal structure of our human reasoning capacities, because such a placement supplies the largest array of clues to help us calculate reliably and to understand and improve the strategic recipe underlying those calculations. We should accordingly respond to a Boltzmannian critic as follows: Of course, we realize that the steel is composed of molecules, but it's better to not employ that kind of language with respect to its higher-scale behaviors, because we want to focus as best we can upon the mathematical ingredients that most effectively align with the behavioral opportunities available at the scale length of large populations of such molecules. When their finer-grained behaviors become relevant, we will shift to replacement submodels better suited to such details, but we should only wish to make those modeling adjustments as circumstances require. Nothing within this sensible "division of linguistic labor" strategy suggests that we don't believe in molecules, but it does explain why we don't explicitly talk about them all of the time. In a similar manner, we can distinguish nature's environmental offerings (how water behaves) from the reasoning policies that promise the greatest rewards of tractability and security in response. At present, the Navier-Stokes equations appear to be the best we can offer in the latter category, but their intrinsically mathematical behaviors remain so uncharted that it's hard to be sure.

In other writings (Wilson 2006), I've identified various ways in which our everyday employment of words like "color" and "hardness" divide into local patchworks that interconnect with one another in allied ways. Any reasonable answer to the philosophical chestnut "to what physical property does the word 'red' correspond?" can reasonably be addressed only if we consider the varying platforms of classificatory opportunity provided within spectral reflection in conjunction with the discriminative capacities of the human eye. And I write "varying platforms" because the relevant informational opportunities frequently vary considerably from material to material and environmental setting to environmental setting. Thus the directional aspects of a metallic object's shiny surface supply considerably richer information about its surface electronic structure than a plank of wood coated with a flat paint (we shall revisit these considerations in Chapter 9 when we consider the utilities of the word "reference"). An adequate unraveling of the complexly entangled "architecture" that governs our everyday "color" discrimination would comprise a significant achievement in "reverse engineering." But this is the patient manner[10] in which I believe that Poynting's challenge should be addressed. We should study how the determinants of practical need and sensory capacity progressively factor a usage into distinct but mutually

[10] Pincock (2010, 120) has characterized my point of view as "patient realism." With respect to color vocabulary, I became so overwhelmed with their complexities that I gave up trying after a point in Wilson (2006). I became subsequently enamored of the admirably developed architectures of modern multiscalar modeling precisely because they effectively illustrate in a controlled manner many of the linguistic developments that I had originally hoped to highlight.

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intercommunicating sectors, whose supportive rationales commonly rest upon significantly different forms of physical underpinning.

In many ways, mathematical and philosophical thought have passed like trains in the night along opposing tracks ("a-whooee, a-whooee"). Both lines of thought initiate within the simple ordinary differential equation (ODE) conception of theoretical content exemplified within the point mass approach to mechanics outlined in Chapter 3. It was within this sheltered mathematical setting that the celebrated French atomist Pierre-Simon Laplace formulated his celebrated articulation of predictive determinism:

> We may regard the present state of the universe as the effect of its past and the cause of its future. An intellect which at a certain moment would know all forces that set nature in motion, and all positions of all items of which nature is composed, if this intellect were also vast enough to submit these data to analysis, it would embrace in a single formula the movements of the greatest bodies of the universe and those of the tiniest atom; for such an intellect nothing would be uncertain and the future just like the past would be present before its eyes.
>
> (Laplace 1814, 4)

In these restricted circumstances, many of the complexities that generate the doctrinal conflicts I have emphasized have not yet clearly emerged. We have not yet witnessed the fruitful but problematic amalgamation of force-based ODEs ($\mathbf{F} = m\mathbf{a}$) with constraint equations that lies at the heart of the tensions that trouble Hertz.[11] Nor have the rich and subtle questions of descriptive harmony that bedevil standard appeals to "boundary conditions" appeared, because "boundary conditions" properly characterized generally appear after applied mathematics shifts into the territory of the partial differential equations (PDE) that are utilized in higher-dimension continuum physics.

Here's a vivid illustration of how these boundary condition complications typically emerge. Suppose we model a metal plate with a suitable set of interior equations (Laplace's equation for a membrane will suffice) and assign it suitable boundary conditions by pushing or pulling on its outer boundary in normal ways. We then confront an escalating set of tensions between interiors and boundaries in the following way. (i) if the boundary is a smooth curve such as an ellipse, interior and boundary demands will accord with one another in the unproblematic matchups that one studies in an introductory class. (ii) Insert some convex corners in the plate, and the previous boundary/interior accord will break down in those locales. But the strain energy within any finite sector will remain finite, so we

[11] More exactly, these combinations were widely utilized, but the complications that constraints introduce with respect to the intuitive existence and uniqueness features of purist ODE settings were not recognized until later in the century.

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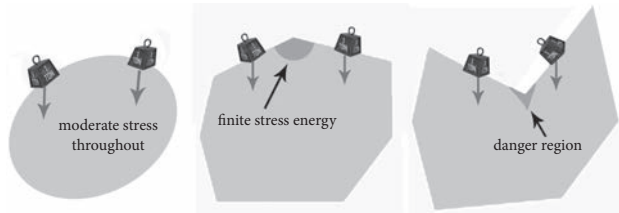

can reformulate our mathematical expectations as to how "solutions" should meet their "boundaries" in a manner that can still be regarded as a suitable account of the energetic situation within the plate. (iii) However, introduce some inset corners into the boundary, and any attempt to supply a viable "solution" will contain patches of strain that blow up to impossible infinities here and there. No real material can sustain strains of that nature. Nonetheless, these same "impossible regions" remain descriptively useful, for real-life cracks generally initiate in these same stress concentrating corners. As such, a liberalized notion of "solution" that tolerates internal infinities can still provide engineers with the warning signposts they require to investigate those particular regions more closely, by consulting a revised modeling that introduces tiny cracks into the strained region's previously unblemished interior (we then determine whether sufficient driving energy is available to elongate these cracks). But to render these determinations mathematically coherent, we must twiddle once again with our notions of "solution" and "boundary accord" (doing so generally demands a fair amount of functional analysis sophistication).

In fact, these are the exact considerations that explain why the subtleties of the "square integrable second distributional derivatives" of which Poston and Stewart complain become so important to the practical engineer. These provide the critical *warning signals* she will need in simulating the behavior of a metal plate upon her computer in a trustworthy manner. This need to search for relevant considerations within a wider stretch of Greater Mathematicsland supplies a further exemplar of the response we offered earlier to Boltzmann.

The conceptual ramifications of these shifting requirements upon interior/boundary accord are nicely captured within a pungent observation from the mathematician Hans Lewy:

> [Mathematical analysis only supplies] kind of hesitant statements. . . . In some ways analysis is more like life than certain other parts of mathematics. . . . There are some fields . . . where the statements are very clear, the hypotheses are clear, the conclusions are clear and the whole drift of the subject is also quite clear. But not in analysis. To present analysis in this form does violence to the subject, in my opinion. The conclusions are or should be considered temporary, also the conclusions, sort of temporary. As soon as you try to lay down exact conditions, you artificially restrict the subject. (Reid 1991, 264)

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These complications comprise some of the mathematical reasons why we should not frame our theory T conceptions of "how science works" according to the simple contours of Newtonian celestial mechanics conducted within a point mass setting (where only the simplest ODE tools are required). As P. G. Tait notes in a passage already cited in Chapter 5:

> [T]he simplicity of the data of the mathematical problem is gone; and physical astronomy, except in its grander outlines, becomes as much confused as any other branch of science. (Tait 1895, 9–10)

We can no longer luxuriate in the unsullied sunshine of the Laplacean mind, but I believe that theory T thinking does exactly that.

However, I have begun to dip deeper into the pool of mathematical technicalities than the central morals of this book require. My guiding intention is merely that of highlighting the characteristic manner in which the tiny confusions and misalignments that everywhere accompany effective description represent subtle tokens of some gradually evolved, but often unnoticed, backgrounded architecture. It is through tacit reliance upon these unnoticed controls that human reasoners manage to extend the applicational ranges of our discourses and correct for their deficiencies. But if these observations are well founded, then the last thing that a philosophical inquirer should wish to do is to smother this vital symptomology beneath an ill-conceived layer of theory T "regimentation."

(vii)

Our chief reason for concentrating upon physics examples in this book lies in the fact that the subject's parochial concerns have served, historically, as the epicenter from which inappropriate conceptions of "rigorous" inquiry have gradually infected other parts of the philosophical corpus. The conceptual clarity that Hertz's axiomatic proposals once promised to the classical physicist continues to serve as the ersatz standard of "rigor" to which a proper "philosophical theory" should piously strive. The fact that genuine conceptual improvements within mechanics have demanded wider-ranging and far messier forms of corrective investigation has mattered little, because the philosophers stopped attending to the original motivating problems. Insofar as I can determine, the wrong turning centrally responsible for coronating this inadequate conception of the diagnostic enterprise as philosophical royalty lies in the mid-twentieth-century manner in which Hertzian axiomatics become rarified into the vagaries of theory T thinking. In so doing, Nagel-like appeals to logic-focused notions such as his versions of "lawlike generalizations" and "initial and boundary conditions" replace the original discriminations actually required within the classifications that remain

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mathematically accurate. Hertz himself is not to blame for these later abuses; his own diagnostic work was truly exemplary. It is merely that he suggested the wrong solution to the tensions that he had correctly noticed.

This deficient circle of logic-like classifications has persuaded most of us that we can happily embrace a schematic vision of "theory" that ignores the very details to which philosophers should most carefully attend, in whatever field they hope to scrutinize. In the previous chapter, we reviewed the manner in which analytic metaphysicians commonly justify their labors as attempts to formulate science-like "theories" of their chosen topics of interest. But this concept of "theory" is entirely mythological in origin, tracing to axiomatic hopes that never materialized).[12] It is within this pseudoscientific vein that the followers of David Lewis justify their speculations. Quoting Theodore Sider once again:

> [M]etaphysicians use standards for choosing theories that are like the standards used by scientists (simplicity, comprehensiveness, elegance, and so on).
>
> (Sider 2007, 6)

It is for these basic reasons that Carnap's "fiction of a completed construction of physics" cannot be tolerated as a "harmless epistemological convenience" but serves as an extirpative presumption that erases the delicate detail required to identify and defang the conceptual obstructions that impede developmental progress. On the basis of a Pollyannish conception of "scientific realism," writers like Sider have concluded that a "completed construction of physics" free of these complex forms of submodeling entanglement will inevitably become available, ready to supply metaphysicians with an uncompromised inventory of the world's "fundamental kinds." With one swift snicker-snack of the vorpal blade, he dispatches all of the troublesome details we have highlighted. But this same tactic vacuums up all of the vital symptomology required to unravel any of the robust mysteries that emerge as the natural collateral damage of any naturally occurring conceptual advance.

For these reasons, we should not transfer Hilbert's dictums on "rigor" uncritically to applied mathematical practice:

> Indeed, the requirement of rigor, which has become proverbial in mathematics, corresponds to a universal philosophical necessity of our understanding; and, on the other hand, only by satisfying this requirement do the thought content and the suggestiveness of the problem attain their full effect. (Hilbert 1902, 441)

[12] This same folklore particularly suits the armchair proclivities characteristic of the "analytic" enterprise. In former times, the term "armchair philosophizing" was regarded as pejorative, but the modern metaphysics school appears to have adopted this designation as a badge of honor. I once attended a conference that proudly featured a comfy chair upon its advertisement poster (and the event was housed in a building dedicated to Emmy Noether, no less).

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This alleged "necessity of our understanding" is cut from the same philosophical fabric as the traditional assumption that the rules of sound methodology are governed by a "universal causal principle" that demands that we must capture nature's behaviors within a fabric of ill-considered "laws of nature." This latter requirement is both vague and patently excessive, and largely stems from the inferential advantages of employing partial differential equations within a wider fabric of corrective feedback modelings. Conducting our researches with full Hilbertian rigor is generally advisable when we pursue goals of a "purely mathematical" character, but these same advantages significantly dissipate when we enter the "indefinite" arena of partial differential equations, where we must frequently decide how to patch sundry shards of mathematized reasoning together in a manner that benefits our understanding of nature. This kind of assembly work may demand a "rigorous" attention to subtle detail, but it is merely the informal rigor of careful consideration that is wanted, not conformity to some dubious standard of preset "universal methodological necessity." As Heaviside puts the point, our first order of business is "to find out what there is to find out" (Heaviside 1912, II 2).

In a nutshell, these observations capture the complaints with respect to "ersatz rigor" with which this monograph began. Internally closed studies of a Hilbertian stripe are frequently quite valuable, but they fail to address a significant layer of further concern with respect to how the individual tools of mathematized calculation should be cobbled together in a manner that befits our hopes of effectively capturing real world behaviors within workable webs of mathematized netting. Occasionally a philosopher might establish a worthy theorem of a Hilbertian character, but these should not be viewed as the primary goals of our profession. The skills we should instead cultivate are those derived from a rich appreciation of how problematic tensions infect a wide variety of otherwise practical forms of discourse, coupled with a thorough acquaintance with the subtle symptomologies of concealed backgrounded architectures. Nothing widely universal can be promised upon a methodological front, because the rewards and punishments of evolutionary development frequently carry their captives to the most astonishing and unexpected locales. But appreciating these widely ranging varieties of remedy can prove useful in itself, for one can feel more secure in "thinking outside the box" if consulting experts are available who have cultivated a cross-disciplinary appreciation of how similar obstacles have been previously surmounted within an array of fields. These consulting skills, I propose, provide a better model for the philosophical enterprise than the attempts to fashion ersatz rigor "theories" based upon armchair experience. This book's chief complaint with respect to contemporary practice is that its currently admired standards have allowed these informal diagnostic skills to shrivel to virtual minimality, rather than comprising the vein of productive suggestion from which an academic philosopher might obtain the rewards of justified satisfaction.

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For the reasons indicated above, I have expressed this book's intended lessons in largely philosophy of science terms. But I firmly anticipate that similar observations apply to other topics of a different stripe. Indeed, J. L. Austin's stated rationale for investigating "excuses" within the context of moral evaluation can be construed as a close copy, mutatis mutandis, of the managerial tasks that interconnected mathematized patchworks perform in sustaining the comparable "mechanisms" of a multiscalar enterprise. He writes:

> [T]o examine excuses is to examine cases where there has been some abnormality or failure: and as so often, the abnormal will throw light on the normal, will help us to penetrate the blinding veil of ease and obviousness that hides the mechanisms of the natural successful act. It rapidly becomes plain that the breakdowns signalized by the various excuses are of radically different kinds, affecting different parts or stages of the machinery, which the excuses consequently pick out and sort out for us. (Austin 1966, 127–8)

In any case, with respect to Sider's appeals to "realism" at the end of the previous chapter, no reasonable formulation of "scientific realism" should demand that future science will suit the schematic dictates of Sider's armchair metaphysics. Here's an analogy to capture the descriptive circumstances we may confront. In the movie *Forbidden Planet*, the fearsome Monster from the Id is invisible, but the crewmen from Starship C-57D can still mark out its positions by refracting tracer bullets along its contours. The Monster is undoubtedly real (it crushes Lt. Farman, inter alia); it is merely that its outline within the tracer contrails is fragmented. And only in this gradualist manner will descriptive inquiries conducted in mathematized terms find themselves able to corral empirical reality. As Planck says, "nature does not allow herself to be exhaustively expressed in human thought" (Planck 1932, 2). Nonetheless, we can improve our descriptive lot considerably if we cleverly link together patches of mathematical technique informed by a wider appreciation of viable recipe.

On the other hand, we must also learn when to abandon a cherished descriptive task when nature churlishly refuses to cooperate with our desires. On these scores, advanced mathematics has reached many negative assessments with respect to inferential tasks that we might wish to accomplish but can't do so feasibly. We might vainly hope to square the circle or construct a perpetual motion device. Or

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establish the stability of the solar system. Or enumerate the highly non-recursive succession of numbered posts that a careening ball might progressively strike within a frictionless pinball machine. Nature will not cooperate with any of these ambitions, and sophisticated mathematical tactics can often tell us why.

(viii)

These negative obstacles represent significant checks upon the more sanguine forms of descriptive optimism that Hertz's axiomatic admirers have unwisely embraced. In Chapter 4, we quoted Ernst Cassirer's unbounded admiration for Hertz's holistic approach to conceptual analysis (leading to a situation where "our minds, no longer vexed, will cease to ask illegitimate questions" (Hertz 1894, 8)). Cassirer concurs in Hertz's recommendation that proper conceptual repair should investigate the practical advantages that a troublesome word like "force" supplies within the full applicational nexus to which it belongs, rather than vainly attempting to isolate some internal rottenness specific to the concept itself. Approaching individual "meaning" in the traditional vein of word-based internal "conceptual analysis" may only unleash the genies of ancestral prejudice that frequently serve as impediments to scientific advance (as Mach frequently emphasizes, earlier stages in word usage never completely die away; they merely become latent, ready to be reinvigorated when the bottle is opened). These are the reasons why Cassirer frequently compares a conceptual arrangement to an integrated "machine" or "organism." Quoting him once again:

> Physical science is a system that must be accepted as a self-contained whole, an organism of which no single part can be made to work without all the others, even those farthest removed, becoming active—some more, some less, but all in some precise degree. (Cassirer 1950, 113)

But a flat descriptive landscape of the type Cassirer seemingly envisions does not necessarily constitute the scientist's *summum bonum*, for the reasons that Tao's "curse of dimensionality" vividly illustrates. We can obtain an excellent appreciation of everything that transpires within a hefty hunk of granite by utilizing a wisely devised multiscalar scheme, despite the fact that the narratives that we spin out in this manner will not unfold smoothly but instead engage in various forms of footnote-justified adjustment pointing in sundry directions (resulting in a narrative closer to *Tristram Shandy* than *McTeague*). Proceeding otherwise may prevent us from seeing the rock but for the grain. The demerits of such an ill-focused enterprise are warmly acknowledged by many of the "realist" scientists we have cited (viz., Planck, Thomson, and Tait).

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As a consequence, the task of an effective reformer is often that of correcting for the inadvertent errors of an established practice while leaving its most vibrant inferential landmarks more or less intact. The result may wind up like an ancient city that has been refitted for modern usage through an elaborate warren of underground tunnels that allow the information stored in building A to flow freely to building B despite the fact that such a communication channel was not envisioned when the city was originally constructed. Throughout this book we have cited the oddly gerrymandered "operational calculus" that the English autodidactic Oliver Heaviside developed in response to the mathematical perplexities of electrical circuits (Wilson 2006, 529–45; Mahon 2017). Through trial-and-error experimentation, Heaviside removed the obstructive barriers within the conventional treatment of differential equations by supplementary "extension elements" (e.g., introducing inverse operations where none had existed before). He cheerfully confessed that he didn't understand the information-processing maneuvers that allow his peculiar "calculus" to reach correct conclusions with their striking efficiencies:

> But then the rigorous logic of the matter is not plain! Well, what of that? Shall I refuse my dinner because I do not fully understand the processes of digestion? No, not if I am satisfied with the result. . . . First, get on, in any way possible, and let the logic be left for later work. (Heaviside 2003, III 9, 461)

But such "later work" is clearly wanted, because Heaviside's modified rules are not completely infallible and rely too heavily upon the "physical intuitions" of the calculator. The task of figuring out from an informational point of view what transpires beneath Heaviside's strange manipulations was bequeathed to a later generation of applied mathematicians. They eventually determined that mathematical formulas that had been formerly parsed as δ/ε point-focused differential equations had shifted to reflect integral equation relationships involving smoothing operations spread over smallish volumes. And they further realized that their reanalysis supplies a better account of the word-to-world relationships that had prompted the use of the original modeling formulas in the first place (Wilson 2017, 324–61).

A similar pattern of revisionary thinking has affected modern classical mechanics practice. The availability of adequate computers originally facilitated the policy of linking submodels together into architectural arrays. The successes obtained by automating intuitive checks-and-balances interconnections in this operational manner subsequently prompted applied mathematicians to subject the rough homogenization policies they employed to a higher degree of rigorous scrutiny.

In these contexts, Heaviside's "logic" signifies a subsequent diagnosis of the underlying "information processing maneuvers" that must roughly underpin the

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majority of inferential transitions permitted within his calculus without necessarily agreeing with all of them. Quite commonly this underlying "logic" proves to be quite different than what the calculus's original employers had once originally fancied (Wilson (2006) describes this as a faulty "semantic picture" of the usage's machinery). I have several times cited a close analogy in our earlier discussions: the card-guessing routines that are easy to implement but whose information-processing strategies demand substantial mathematical excavation. To protect against applicational error, we generally want to study how information is passed along the reasoning routines permitted within the novel calculus, in part to gain a sharper appreciation of where such routines are likely to lead to error. Computer scientists dub these informational tracings as "verifications of correctness," whereas logicians label them as "soundness proofs." These represent the after-the-fact investigations we will wish to pursue after we have roughed out the network of inferential connections at play within a practical usage. We will want to understand the objective landscape that secretly informs the successes of an evolved calculus such as Heaviside's, without ratifying every claim that it certifies. To repair its faults, we should endeavor "to follow the information" and repair the unavoidable glitches that will have inevitably crept in. In doing so, a variety of somewhat different reconstructions may prove viable (Heaviside's calculus has been "rigorized" in a variety of ways). Such variations with respect to the improvement of an inductively established reasoning recipe are to be expected and reflect the manners in which such practices have gradually solidified from a more amorphous set of practical skills. Biological evolution rarely produces perfectly designed specimens, and the same moral applies to linguistic development as well.

Wilson (2006) dubs these successive stages in linguistic reform as "seasonalities" whose characteristic phenomenologies we shall explore a bit further in Chapter 9. The central moral with respect to our present concerns resides in the observation that it is exactly along these lines of "seasonal" exploration that a scientific realist should expect to uncover the considerations that distinguish the "scaffolding" aspects of a successful descriptive practice from the objective opportunities that support it.

Sider's version of "realism" does not proceed in this manner at all, but instead awaits the hypothetical "fundamental physics" that Sider assures us is someday forthcoming.[13] From its nicely regimented contours he believes that an eager metaphysician will be able to extract its "ontological commitments" in Quine's recommended manner, bypassing all of the messy complications upon which I insist. How have Sider's utopian expectations managed to plant such deep

[13] Even for "theories" like classical mechanics that eventually lose the guiding imprimatur of nature. Sider appears willing to appeal to "the possible models of classical mechanics" in the absence of any evident set of axioms that might successfully carve out such structures.

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roots within our academic psyches? How has a proposal for methodological reform that didn't work (Hertz's axiomatization) turned into a "convenient idealization" that philosophers can confidently embrace within their methodological musings? Why has philosophy of science largely abandoned the careful distinctions of the applied mathematicians in favor of the wan schematisms and ill-defined "laws of nature" characteristic of theory T thinking?

(ix)

Let us now review several further incidents that have contributed to this strange transmogrification. Recall Hertz's and Helmholtz's unfortunate contrast between the "gay conceptual garments" that enfold the mental conceptions (Hertz's "images") that we vibrantly grasp and the dimmer exterior properties that we can only consider in the skeletal manner of a Harpo-imitates-Groucho correspondence.[14] Let's call the vocabularies whereby we reach out to these minimally grasped properties-in-themselves as "theoretical vocabulary" V_T. Influenced by Hertz and Hilbert,[15] philosophers such as Moritz Schlick conceived the radical ambition that an adequately closed axiomatic system **A** might completely settle the truth-values of sentences that contain these "theoretical" V_T terminologies through "implicit definition": as soon as the extensions of the observational predicates within T become determined, **A**'s axiomatic demands will force unique truth-values upon the V_T remainder. If so, we needn't worry further about word-to-world underpinnings of such words because their contributions to sentential truth-values can be estimated from **A**-internal considerations alone. Such hopes frame the motivational basis for the brash manifesto cited earlier:

> Philosophy is to be replaced by the logic of science—that is to say, by the logical analysis of the concepts and sentences of the sciences, for the logic of science is nothing other than the logical syntax of the language of science.
>
> (Carnap 1934, xiv)

Fulfilling this program encountered many familiar snags in the years to follow, but hazy doctrines of an "implicit definability" character live on to this day, including Thomas Kuhn's notorious contention (Kuhn 1962) that paradigm-dependent "conceptual incommensurabilities" prevent successive scientific

[14] Worrall (1989) strikes me as cut from this same conceptual cloth.

[15] Hilbert (1925) hopes that standard objections to the extension elements within geometry and infinitary quantifications within logic might be painlessly avoided in this hermetically enclosed syntactical manner.

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generations from comprehending one another.[16] Shortly thereafter, Hilary Putnam hoisted a banner of self-proclaimed "scientific realism" in which he reasonably objected to Kuhn's extreme position on the grounds that scientific dialectics actually concern a common referential basis upon which all parties actually agree, whether the "theories" they endorse are similar or not (Putnam 1975). In particular, Putnam contended that the disputants in controversies concerning, e.g., "temperature" were all concerned with the focal characteristic of "possessing an averaged molecular kinetic energy of fixed value E" without realizing that they were doing so. Based upon this de facto identification of referential target, we can meaningfully assess which of the parties had advanced correct claims with respect to "temperature" and which had not.

Let us here note that historians of science commonly employ allied referential idioms of "reference" in identifying a central objective focus within a scientific practice. In this vein, we can usefully declare that "When Thomas Young spoke of 'monochromatic light,' he was actually *referring* to a power spectrum decomposition of a statistical steady state ensemble stretching over a sizable temporal interval encompassing a large number of pulse-sourced events." In doing so we can simultaneously acknowledge the important lineage of optical research that Young initiated as well as the surprising revisions in our conception of "spectrum" that became later recognized. As the mathematician Robert Woodhouse once wrote, "a method which leads to true results must have its logic":

> Now if operations with any characters or signs lead to just conclusions, such operations must be true by virtue of some principle or other; and the objections against the [facilitating quantities] ought to be diverted upon the unsatisfactory explanation given of their nature and uses. (Woodhouse 1802, 90)[17]

This is the same sense of "logic" that we have encountered in Heaviside: an unraveling of the mysterious strategic patterns that explicate how input information becomes secretly processed to effective ends within a successful descriptive practice. As we have already noted, practitioners at sundry developmental stages within a usage's evolving career frequently entertain wildly inaccurate conceptions of the informational processes at issue, in the manner of an amateur magician who believes that he can really read minds. Sympathetically evaluating the typical motley of partially just and partially unjust conclusions that characterize the historical record often proves context sensitive in extremely irregular ways.

[16] Kuhn's exact reasons remain rather inscrutable due to his reliance upon the murky contours of his "paradigms." Other authors have attributed allied "incommensurabilities" to framework disagreements conceptualized in Carnap's general manner.

[17] I first learned of this passage from Smith (1851). Woodhouse's original essay is more conservatively intended than Smith's citation suggests, which isn't surprising in view of the radical reshapings that mathematics underwent in the half century that separates these authors.

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Nonetheless, identifying their proximate objective supports can prove enormously enlightening in appreciating the unexpected twists and turns that our evolving descriptive policies evidence as they gradually wend their ways to improved performance. I believe that these adjustments represented Putnam's original focal concern. I was a thesis student of his and was greatly inspired by the suggestion that the history of science might serve as an excellent laboratory for directly studying the ways in which concrete word-to-world relationships progressively adjust and refine themselves under the goad of practical achievement. Because the resulting word-to-world discriminations commonly prove very complicated, some parties have suggested that the non-trivial evaluative uses of "reference" should be jettisoned altogether (Field 2001). I thoroughly reject such defeatism. Yes, the byways of language are often byzantine, but that's simply an important aspect of the natural history of *homo sapiens* with which we must contend in any useful form of "philosophizing." I shall append a few supplementary remarks on the value of diagnosing non-trivial word-to-world relationships in terms of a semantically liberated notion of "reference" in our final chapter.

For reasons I never fully understood, Putnam abandoned these straightforward "study conceptual development within the history of science" policies in favor of a rigid form of alleged "scientific realism" that he articulared (in company with Richard Boyd) as a developmental slogan "The terms in a mature science typically refer" (Putnam 1978, 21). This thesis anticipates that the development of any worthy body of scientific doctrine will eventually reach a condition in which its vocabularies will subscribe to the simple "Fido"/Fido alignments of traditional semantic expectation. Putnam labeled these hypothesized correlates as "natural kinds or properties," selecting as his chief paradigm the alleged fact that "possesses a fixed degree of averaged kinetic energy" serves as the "natural kind" to which the term "temperature" eventually gravitates. This enormously influential thesis has become popularly known as "reference magnetism" and is presumed to follow from the alleged truism that "Of course, scientists will want their terminologies to eventually possess unambiguous references." Metaphysically inspired authors have revived antiquated conceptions of scientific methodology by suggesting that Putnam's focusing processes indicate that one of science's primary goals is that of seeking the hidden *essences* of natural states of affairs. Such mystical claims would have sent chills down the spines of our hard-headed nineteenth century avatars.

On this score, let us simply observe that this characterization of "temperature's" evolving career represents a philosopher's myth pure and simple. If it were correct, we couldn't attribute a "temperature" to an ordinary rubber band. But I won't enlarge upon these details here.[18] In any case, Putnam's attempt to codify the

[18] See Wilson (2017, 190–2). The motivational complications are intimately connected with the energetic cascades discussed in Chapter 6.

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expectations of "scientific realism" in *linguistic terms* should have been rejected as inappropriate from the get-go (there's none of that in Max Planck, for example).

Be this as it may, Putnam's futuristic construal of "scientific realism" in "natural kind" terms opened the floodgates to a revived conception of "metaphysics" that imposes prohibitory restrictions upon scientific innovation closely akin to the tenets that our scientific avatars feared.[19] David Lewis's writings (Lewis 1983) represents a critical stage in these developments, and Sider's remarks follow him closely. Observe how the latter appeals to "scientific realism" in the programmatic declaration that I cited at the end of the previous chapter:

> A realistic picture of science leaves room for a metaphysics tempered by humility.
> (Sider 2007, 6)

It is Putnam's linguistic "realism" that is here invoked, not one grounded in the direct consideration of real-life developmental process. For Sider, "realism" demands the eventual production of a "fundamental science" firmly planted upon univocal "natural property" foundations. As already indicated, he shares the conviction that, although we do not possess such an admirable formalism at present, one day we surely will. Because he further presumes that this hypothetical science will conform to the convenient-for-philosophy contours of single-leveled traditional theory T expectation, Sider believes he can safely ignore the "small metaphysics" complexities of ascertaining the objective supports that sustain the utilities of our current employments of scientific vocabularies. Through these swift "idealizations," Sider leapfrogs past all of the inconvenient complexities we have noted in these pages, allowing him to luxuriate in a rosier future free of such concerns. Along the way our philosopher/scientists' original misgivings with respect to "essences," "natural kinds," and allied forms of strong "metaphysical" demand appear to have become utterly forgotten. My suggestion, of course, is that we should attend more warmly to their wise councils.

Although Sider portrays his future-based doctrines as ones of "modest expectation," they are not modest at all in the sense that he feels entirely certain of the based categories that must be completed within this hypothesized "fundamental science" (the precise future occupants of these firm categories strike Sider as matters of comparatively little metaphysical interest). That assumption granted, philosophers can busily occupy themselves in constructing a pre-scientific

[19] I find it striking that, despite their promises of remaining up to date with science, a considerable segment of the "analytic metaphysics" literature actually struggles with venerable problems within classical mechanics. And they commonly do so with mathematical tools long recognized to be inadequate to these purposes (Smith 2007; Wilson 2008). I once gave a lecture in which I mischievously posted an aerial photograph of a prominent university, pointing out the locations of the departments of philosophy and fluid mechanics. I then asked, "Don't these folks talk to one another?"

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"theory" of metaphysical requirement by adhering to the same methodological standards that allegedly characterize rigorous "theory" construction within any designated field. In this imitative fashion, Sider reinstates an essentially apriorist conception of the metaphysical enterprise as a discipline that delineates the contours that any respectable science must later strive to instantiate, largely based upon one's armchair intuitions as to how the required categories of "object," "property," "law," and "cause" might be filled in.

But no reasonable construal of "scientific realism" can promise Sider an all-encompassing "fundamental physics" of the sort he anticipates. Quite the contrary; an objective assessment of the limits that constrain the reach of our available mathematical tools may identify descriptive lacunae that can only be bridged through means other than a single-leveled mathematized formalism. A gambler may huff and puff all he likes, but he won't be able to successfully augur the outcomes of a severely unstable gaming device. And the processes of energetic decoherence in nature may demand similar descriptive liberties.[20]

But in the breezy manner of typical "theory T" appeal, the subtle symptomatology that I have defended as essential to serious conceptual rectification has been cast aside as unimportant. Once these dismissive proclivities take hold, they are extremely hard to dislodge. In my personal experience, I have been dumbfounded by the almost complete indifference to real-life example among Sider's analytic metaphysics cohort: "Oh those doctrinal peculiarities may seem troublesome now, but I needn't worry about any of them. Your worries are merely epistemological." Any suggestion that pursuing "metaphysics" might require a diagnostic probing of linguistic architecture strikes them as a betrayal of the designated mission.

Such are the downstream consequences of the wrong turning that analytic endeavor took in the 1930s, prompted by Carnap's eager embrace of Hertz's hope that confusions of classical mechanics might be remedied through a stiff dose of axiomatic medicine. The fact that the subject did not respond entirely favorably to the treatment occasioned no subsequent alarms within the philosophical community, largely because in the meantime the quantum physicists had lost interest and shunted the patient off to the wards of the departments of engineering. Hertz's and even Carnap's specific proposals have been long forgotten within our contemporary context, and most modern metaphysicians would be highly offended to be accused of seeking axiomatizations in Carnap's fuddy-duddy manner. But these same writers have evidently forgotten virtually all of

[20] I might further mention that there is a certain sector of contemporary philosophers of quantum mechanics who refuse to consider any aspect of their subject unless it has been properly axiomatized (for a critique, see Williams (2019)). Like Casey at the bat, they disdain all irregular features such as renormalization and far field asymptotics as "that ain't my style" as the orbs whiz across their plates. They might recall that the mighty Casey struck out through stern adherence to these scruples.

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the significant methodological concerns with which our nineteenth century predecessors forebears actively struggled, in their attempts to liberate their improving disciplines from the ill-considered manacles of armchair philosophizing. But these same misgivings remain as relevant today as they were back then. I can't see how contemporary analytic effort could have possibly returned to its present-day forms of excessive "metaphysical" presumption without the intervening amnesia of theory T fogginess. And the surface allure of "ersatz rigor" can't obliviate the fact that important considerations have become forgotten along the way. By all means, we should continue to reflect upon our relationships with the exterior world in a "metaphysical" vein, but we should also keep a closer eye upon the range of smaller concerns that generate the grander extravaganzas.

Thus the "alternative history" of analytic philosophy I would have liked to have witnessed would have returned to Mach's and Hertz's struggles with conceptual disharmonies and built upon their recommendations in a more nuanced manner, based upon a careful study of the successful stratagems that we encounter within real-life deductive practice. We have remembered only the dubious remedy, not the informed diagnosis.

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